KevaBook Requirements Specification

1. Introduction

1.1 Project Overview

KevaBook is a cloud-based booking platform designed to simplify appointment scheduling and booking management for freelancers, tutors, consultants, coaches, and small service-based businesses.

The platform enables business owners to create professional business profiles, define the services they offer, manage their availability, and receive bookings from clients through a centralized online platform. Clients can discover businesses, view available services, and book appointments without the need for manual scheduling through phone calls, emails, or messaging applications.

KevaBook follows a multi-tenant model, allowing multiple independent businesses to operate within the same platform while maintaining their own business information, services, schedules, and bookings.

The system is intended to reduce administrative overhead for service providers, improve booking efficiency, and provide clients with a convenient self-service booking experience.

1.2 Problem Statement

Many freelancers and small service-based businesses rely on manual appointment scheduling through phone calls, emails, social media messages, or messaging applications such as WhatsApp. These approaches often lead to scheduling conflicts, missed appointments, double bookings, and inefficient communication between businesses and clients.

Additionally, many small businesses lack the resources to develop and maintain their own booking systems.

KevaBook addresses these challenges by providing a centralized platform where businesses can manage their services and appointments while allowing clients to book available time slots online.

1.3 Project Goals

The primary goals of KevaBook are:

- Simplify appointment scheduling for service-based businesses.
- Enable businesses to manage services and availability online.
- Allow clients to book appointments through a self-service platform.
- Reduce scheduling conflicts and double bookings.
- Automate booking notifications and reminders.
- Provide a scalable foundation for future growth and feature expansion.

1.4 Target Users

The platform targets the following users:

Business Owners

Business owners are individuals or organizations that provide services and wish to accept bookings online.

Examples include:

- Tutors
- Consultants
- Freelancers
- Coaches
- Therapists
- Accountants
- Photographers
- Trainers

Clients

Clients are individuals seeking services from businesses registered on the platform.

Clients can:

- Browse available services
- View business profiles
- Book appointments
- Receive booking confirmations and notifications

1.5 Scope of MVP

The Minimum Viable Product (MVP) will focus on the core booking workflow.

Included Features:

- User registration and authentication
- Business profile management
- Service management
- Availability management
- Appointment booking
- Booking management
- Email notifications
- Sms notifications

Excluded from MVP:

- Online payments
- Ratings and reviews
- Video conferencing integration
- Calendar synchronization
- Analytics dashboards
- Team/staff management
- Subscription billing

2. Functional Requirements

2.1 User Management

FR-01 User Registration

The system shall allow users to create an account using an email address and password.

FR-02 User Authentication

The system shall allow registered users to log in using valid credentials.

FR-03 Password Reset

The system shall allow users to reset their password through email verification.

FR-04 User Profile Management

The system shall allow users to view and update their profile information.

2.2 Business Management

FR-05 Business Profile Creation

The system shall allow authenticated users to create a business profile.

FR-06 Business Profile Update

The system shall allow business owners to update business information.

Business information may include:

- Business name
- Description
- Contact details
- Location

- Business logo

FR-07 Business Profile Viewing

The system shall allow clients to view business profiles.

2.3 Service Management

FR-08 Service Creation

The system shall allow business owners to create services.

Each service shall contain:

- Service name
- Description
- Duration
- Price
- Category

FR-09 Service Update

The system shall allow business owners to update service details.

FR-10 Service Removal

The system shall allow business owners to deactivate or remove services.

FR-11 Service Listing

The system shall allow clients to view available services offered by a business.

2.4 Availability Management

FR-12 Availability Configuration

The system shall allow business owners to define working days and working hours.

FR-13 Availability Update

The system shall allow business owners to modify their availability schedule.

FR-14 Slot Generation

The system shall automatically generate available booking slots based on defined availability and service duration.

2.5 Booking Management

FR-15 Service Booking

The system shall allow clients to book an available service.

FR-16 Booking Confirmation

The system shall generate a booking record after successful booking.

FR-17 Booking Viewing

The system shall allow business owners to view bookings associated with their business.

FR-18 Booking Cancellation

The system shall allow bookings to be cancelled.

FR-19 Double Booking Prevention

The system shall prevent multiple bookings from being created for the same time slot.

FR-20 Booking Status Management

The system shall maintain booking statuses including:

- Pending

- Confirmed
- Cancelled
- Completed

2.6 Search and Discovery

FR-21 Business Search

The system shall allow clients to search for businesses.

FR-22 Service Search

The system shall allow clients to search for services.

FR-23 Business Service Viewing

The system shall allow clients to browse services offered by a selected business.

2.7 Notification Management

FR-24 Booking Confirmation Notification

The system shall send an email and sms notification when a booking is successfully created.

FR-25 Booking Cancellation Notification

The system shall send an email and sms notification when a booking is cancelled.

FR-26 Reminder Notification

The system shall send appointment reminder notifications before scheduled appointments.

3. Non-Functional Requirements

3.1 Performance

NFR-01 Response Time

The system shall process and respond to 95% of API requests within 500 milliseconds under normal operating conditions.

NFR-02 Booking Processing Time

The system shall create and confirm bookings within 1 second under normal operating conditions

NFR-03 Concurrent Users

The system shall support at least 100 concurrent users while maintaining the response time requirements defined in NFR-01 and NFR-02.

3.2 Scalability

NFR-04 Horizontal Scalability

The system shall support horizontal scaling of services as demand increases.

NFR-05 Service Independence

System components shall be deployable independently to support future expansion.

3.3 Availability

NFR-06 System Availability

The platform should maintain high availability and minimize service interruptions.

NFR-07 Fault Isolation

Failures in one service should not cause the entire platform to become unavailable.

3.4 Security

NFR-08 Authentication

Access to protected resources shall require user authentication.

NFR-09 Authorization

Users shall only access resources they are permitted to manage.

NFR-10 Data Protection

Sensitive information shall be protected during transmission and storage.

NFR-11 Password Security

User passwords shall be stored using secure hashing algorithms.

3.5 Reliability

NFR-12 Data Consistency

The system shall maintain consistent booking records.

NFR-13 Booking Integrity

The system shall ensure that booking transactions are processed accurately and without duplication.

NFR-14 Availability Accuracy

The system shall prevent double booking of appointments slots and maintain booking consistency under concurrent booking requests.

3.6 Maintainability

NFR-15 Modular Design

The system shall follow a modular microservices architecture to facilitate maintenance and future development.

NFR-16 Code Quality

The system shall follow established coding standards and development best practices.

3.7 Usability

NFR-17 User-Friendly Interface

The platform shall provide an intuitive and easy-to-use interface for both business owners and clients.

NFR-18 Mobile Accessibility

The platform shall be accessible through modern desktop and mobile web browsers.

3.8 Observability

NFR-19 Logging

The system shall maintain application logs for monitoring and troubleshooting.

NFR-20 Monitoring

The system shall provide health monitoring for deployed services.

NFR-21 Distributed Tracing

The platform should support distributed request tracing across microservices.

4. Use Case Analysis

4.1 Actors

Business Owner

A registered platform user who provides services and manages bookings through the platform.

Responsibilities:

- Create and manage a business profile
- Create and manage services
- Configure availability schedules
- View bookings
- Cancel bookings
- Receive booking notifications

Client

A person seeking services offered by businesses registered on the KevaBook platform. Clients are not required to create an account before booking appointments.

Responsibilities:

- Search for businesses
- View business profiles

- View available services
- Select available appointment slots
- Book appointments
- View booking details through booking confirmations
- Cancel appointments using a booking reference or secure cancellation link

System

The automated platform responsible for managing bookings and supporting platform operations.

- Authenticate business owners
- Generate available booking slots
- Validate booking requests
- Prevent double bookings
- Manage booking statuses
- Send notifications and reminders

4.2 Business Owner Use Cases

UC-01 Register Account

Description

A business owner creates a new account on the platform.

Preconditions

- The business owner does not already have an account.
- The provided email address is not associated with an existing account.

Main Flow

1. Business owner accesses the registration page.
2. Business owner enters registration information..
3. System validates the submitted information.
4. System creates the account.
5. System confirms successful registration.

Postconditions

- A business owner account is successfully created.
- The business owner can log in to the platform.

UC-02 Login

Description

A registered business owner logs into the platform to access and manage their business information, services, availability, and bookings.

Preconditions

- A business owner account exists.
- The business owner has valid login credentials.

Main Flow

1. Business owner navigates to the login page.
2. Business owner enters email and password.
3. System validates the credentials.
4. System authenticates the business owner.
5. System grants access to the platform.

Postconditions

- The business owner is authenticated.
- The business owner can access protected platform features.

UC-03 Create Business Profile

Description

Business owner creates a business profile.

Main Flow

1. Business owner enters business information.
2. System validates the submitted data.
3. System creates and persists the business profile.
4. System links the business profile to the business owner account.

Postconditions

- A business profile is successfully created and associated with the business owner.

UC-04 Manage Services

Description

A business owner creates, updates, or deactivates services offered by their business.

Main Flow

1. A business owner accesses service management.
2. A business owner creates or updates a service.
3. The system validates service data.
4. The system saves changes to the service catalog.
5. The system associates service with the business profile.

Postconditions

- Service data is persisted and associated with the correct business.

UC-05 Configure Availability

Description

A business owner defines working hours and availability rules for booking.

Postconditions

- Business owner is authenticated.
- Business profile exists.

Main Flow

1. The business owner selects availability settings.
2. The business owner defines a schedule, like working hours and days, and optional breaks.
3. The system validates schedule configuration
4. The system stores availability rules

Postconditions

- Available booking slots can be generated.

UC-06 View Bookings

Description

Business owner views bookings associated with their business.

Main Flow

1. Owner requests booking list.
2. System retrieves bookings.
3. System displays bookings.

Postconditions

- Booking information is visible.

UC-07 Cancel Booking

Description

Business owner cancels an existing booking.

Main Flow

1. The business owner selects a booking.
2. The system validates that the booking can be cancelled.
3. The system updates booking status to **Cancelled**.
4. The system triggers cancellation notification to the client.

Postconditions

- Booking status is updated to Cancelled.
- Client is notified of cancellation.

4.3 Client Use Cases

UC-08 Search Businesses

Description

Client searches for businesses.

Main Flow

1. Client enters search criteria.
2. System returns matching businesses.

Postconditions

- Matching businesses are displayed.

UC-09 View Business Profile

Description

Client views business information.

Main Flow

1. Client selects a business.
2. The system displays profile details.

Postconditions

- Business profile is displayed.

UC-10 View Services

Description

Client views services offered by a business.

Main Flow

1. Client selects business.
2. System retrieves services.
3. Services are displayed.

Postconditions

- Service list is displayed.

UC-11 Book Appointment

Description

Client books a service appointment.

Preconditions

- Service exists.
- Time slot is available.

Main Flow

1. Client selects service.
2. Client selects date.
3. Client selects a time slot.
4. The system validates availability.
5. The system creates booking.
6. Confirmation notification is sent.

Postconditions

- Booking is created.

UC-12 Cancel Appointment

Description

Client cancels an existing booking.

Main Flow

1. Client selects booking.
2. Client requests cancellation.
3. System updates booking status.
4. Notification is sent.

Postconditions

- Booking status becomes Cancelled.

5. Domain Analysis

The domain model defines the core business entities and relationships of the KevaBook platform. It is designed to support a multi-tenant booking system where business owners manage services and availability, while clients book appointments without requiring an account.

5.1 Business Owner

Represents a registered platform user who owns and manages a business.

Attributes

- OwnerId
- FirstName
- LastName
- Email
- Password
- CreatedAt
- UpdatedAt

Relationships

- A Business Owner may own one or more Businesses
- A Business belongs to one Business Owner

5.2 Business

Represents a service-based business registered on the platform.

Attributes

- BusinessId
- Name
- Description
- ContactEmail
- PhoneNumber
- LogoUrl
- CreatedAt
- UpdatedAt

Relationships

- Belongs to one Business Owner
- Offers many Services
- Has many Availability rules
- Has many Bookings

5.3 Service

Represents a bookable service offered by a business.

Attributes

- ServiceId
- Name
- Description
- Duration
- Price
- Category
- isActive

Relationships

- Belongs to one Business.
- Can be booked many times.

5.4 Availability

Represents the working schedule and booking rules of a business.

Attributes

- AvailabilityId
- BusinessId
- DayOfWeek
- StartTime
- EndTime

Relationships

- Belongs to one Business.

5.5 Booking

Represents an appointment between a client and a business.

Attributes

- BookingId
- ServiceOfferId
- ClientName
- ClientEmail
- ClientPhoneNumber
- BookingDate
- StartTime
- EndTime
- Status (Pending, Cancelled, Confirmed, Completed)
- CreatedAt
- CreatedAt
- UpdatedAt

Relationships

- References one Service.

- References one Business.

5.6 Notification

Represents a notification event.

Attributes

- NotificationId
- Recipient
- Type (Email, Sms)
- Subject
- Message
- Status (Pending, Sent, Failed)
- SentAt
- CreatedAt

Relationships

- May be associated with a Booking.

6.1 No Client Entity

There is intentionally NO Client entity because:

- Clients do not authenticate
- Clients are transient users
- Client identity is embedded in Booking

This reduces complexity and improves conversion rate.

6.2 Booking is a Core Aggregate

Booking is the **most important entity in the system** because it:

- Connects Business
- Connects Service
- Contains Client data
- Drives Notifications
- Enforces availability rules

6.3 Business is the Main Tenant Boundary

Everything is scoped under Business:

- Services belong to Business
- Availability belongs to Business
- Bookings belong to Business

This defines your **multi-tenant architecture boundary**.

6.4 Notification is Event-Driven

Notifications should NOT be tightly coupled to Booking.

Instead:

- Booking created → event emitted
- Notification service consumes event